

A Tale of Two Headers: A Formal Analysis of Inconsistent Click-Jacking Protection on the Web

A Tale of Two Headers: A Formal Analysis of Inconsistent Click-Jacking Protection on the Web - Stefano Calzavara, Sebastian Roth, Alvisè Rabitti, Michael Backes, Ben Stock, [usenix.org](https://www.usenix.org).

- Published: date not stated
- Original: <https://www.usenix.org/conference/usenixsecurity20/presentation/calzavara>
- Preserved from: <https://www.usenix.org/conference/usenixsecurity20/presentation/calzavara> (stored) on 2026-08-11
- Capture timestamp: 20200813181347
- Licence: unknown

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Abstract:

Click-jacking protection on the modern Web is commonly enforced via client-side security mechanisms for framing control, like the X-Frame-Options header (XFO) and Content Security Policy (CSP). Though these client-side security mechanisms are certainly useful and successful, delegating protection to web browsers opens room for inconsistencies in the security guarantees offered to users of different browsers. In particular, inconsistencies might arise due to the lack of support for CSP and the different implementations of the underspecified XFO header. In this paper, we formally study the problem of inconsistencies in framing control policies across different browsers and we implement an automated policy analyzer based on our theory, which we use to assess the state of click-jacking protection on the Web. Our analysis shows that 10% of the (distinct) framing control policies in the wild are inconsistent and most often do not provide

any level of protection to at least one browser. We thus propose recommendations for web developers and browser vendors to mitigate this issue. Finally, we design and implement a server-side proxy to retrofit security in web applications.

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BibTeX

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@inproceedings {251564, author = {Stefano Calzavara and Sebastian Roth and Alvise Rabitti and Michael Backes and Ben Stock}, title = {A Tale of Two Headers: A Formal Analysis of Inconsistent Click-Jacking Protection on the Web}, booktitle = {29th {USENIX} Security Symposium ({USENIX} Security 20)}, year = {2020}, isbn = {978-1-939133-17-5}, pages = {683--697}, url = {https://www.usenix.org/conference/usenixsecurity20/presentation/calzavara}, publisher = {{USENIX} Association}, month = aug, }
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